PARISHRAM 2.0

CLASS 12TH 2026

Basic Math and Calculus

PHYSICS One-Shot

BY-DEV AGNANI SIR



Topics to be covered:

1. Algebra
2. Mensuration
3. Trigonometry
4. Logarithms
5. Differential Calculus
6. Integral Calculus

(7.) Some important Graphs

(8.) Vectors.

Algebra

Common Identities

→ $(a+b)^2 = a^2+b^2+2ab$

→ $(a-b)^2 = a^2+b^2-2ab$

→ $(a^2-b^2) = (a+b)(a-b)$

ex: $5^2-4^2 = (5+4)(5-4)$

$= 9 \times 1 = \boxed{9}$.

ex: $(101)^2 = (100+1)^2$

$= 10000 + 1 + 2\times100\times1$

$= \boxed{10,201}$

ex: $(99)^2 = (100-1)^2$

$= 10000 + 1 - 2\times100\times1$

$= \boxed{9,801}$.

Algebra

Quadratic Equations

$$ax^2 + bx + c = 0$$

- x is the Root of the equation.
- a, b, c are the constants.
- Degree of the eqⁿ = Max. Power of x.

Degree = 2 → We'll have two roots for this eqⁿ.

Values of x

Shri Dharacharya Formula :-

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \qquad x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

QUESTION

Solve the equation: $6x^2 - 13x + 6 = 0$

$36 = 6 \times 6$
$36 = 4 \times 9$
$36 = 9 \times 4$
$36 = 3 \times 12$
$36 = 12 \times 3$

→ By SDF :-

$$6x^2 - 13x + 6 = 0$$

$a = 6,\ b = -13,\ c = 6$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow x = \frac{13 \pm \sqrt{25}}{2(6)}$$

$$\Rightarrow x = \frac{13 \pm 5}{12}$$

$x = \frac{3}{2}$

$x = \frac{2}{3}$

→ By Middle-term Splitting :-

$$6x^2 - 13x + 6 = 0$$

$$\Rightarrow 6x^2 - 9x - 4x + 6 = 0$$

$$\Rightarrow 3x(2x - 3) - 2(2x - 3) = 0$$

$$\Rightarrow (3x - 2)(2x - 3) = 0$$

$3x - 2 = 0$

$x = \frac{2}{3}$

$2x - 3 = 0$

$\Rightarrow x = \frac{3}{2}$

Algebra

Binomial Theorem

$x \longrightarrow$ Real number

$n \longrightarrow$ +ve no., -ve no., fraction.

$$(1+x)^n = 1 + nx + \frac{n(n-1)x^2}{2!} + \frac{n(n-1)(n-2)x^3}{3!} + \ldots\ldots\ldots$$

$2! = 2$ Factorial $= 2 \times 1$

$3! = 3 \times 2 \times 1$

$5! = 5 \times 4 \times 3 \times 2 \times 1$

$n! = (n)(n-1)(n-2) \ldots\ldots 3 \times 2 \times 1$

QUESTION

The acceleration due to gravity at a height h above the surface of the earth (radius = R) is given by :-

$$g' = \frac{gR^2}{(R+h)^2}$$

If $h << R$, then show that

$$g' = g\left(1 - \frac{2h}{R}\right)$$

Sol. $g' = \dfrac{gR^2}{(R+h)^2} = \dfrac{gR^2}{\left(R + \frac{hR}{R}\right)^2} = \dfrac{gR^2}{\left\{R\left(1 + \frac{h}{R}\right)\right\}^2} = \dfrac{gR^2}{R^2\left(1 + \frac{h}{R}\right)^2}$

$\left\{\begin{array}{l} h << R \\ h^2 << h << R^2 \end{array}\right\}$

$\Rightarrow g' = g\left(1 + \frac{h}{R}\right)^{-2} = g\left[1 + (-2)\frac{h}{R} + (\quad)\left(\frac{h}{R}\right)^2 \cdots\cdots\right] \Rightarrow \boxed{g' = g\left(1 - \frac{2h}{R}\right)}$

Mensuration

Important Formulae

#Areas :-

→ Cube = $6a^2$

→ Cuboid = $2(lb+bh+hl)$

→ Cylinder (CSA) = $2\pi rh$

↳ (TSA) = $2\pi rh + 2\pi r^2$

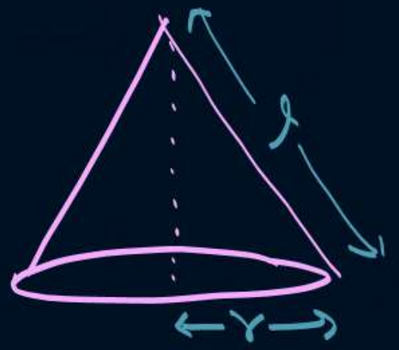
$= 2\pi r(h+r)$

→ Sphere = $4\pi r^2$

→ Cone (CSA) = πrl

(TSA) = $\pi rl + \pi r^2$

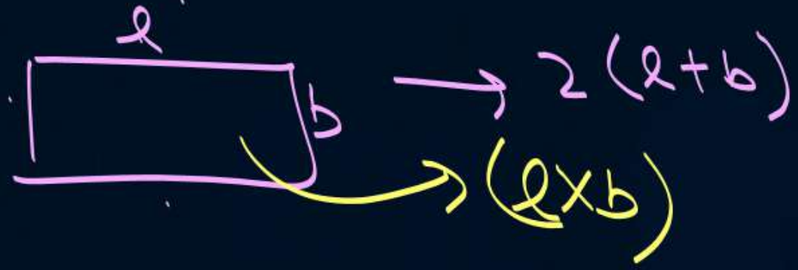
$= \pi r(l+r)$

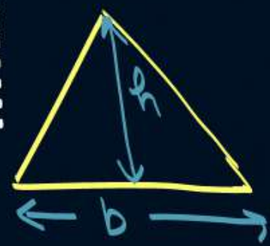




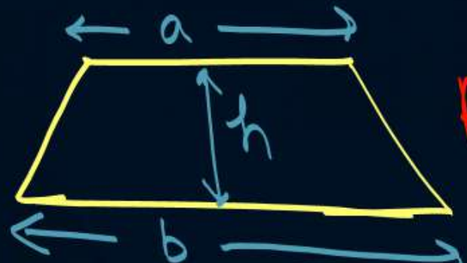








Area = $\frac{1}{2} \times b \times h$



Area = $\frac{1}{2} \times h \times (a+b)$

Trapezium.

Mensuration

Important Formulae

Volume :-

→ Cube = a^3

→ Cuboid = $l \times b \times h$

→ Sphere = $\frac{4}{3}\pi r^3$

→ Cylinder = $\pi r^2 h$

→ Cone = $\frac{1}{3}\pi r^2 h$

Que → We have a cylinder and a cone of same height ($h = 2m$) and same radius ($r = 0.5\,m$). Find the difference between the amount of water they can hold.

Trigonometry

Pythagoras Theorem :-

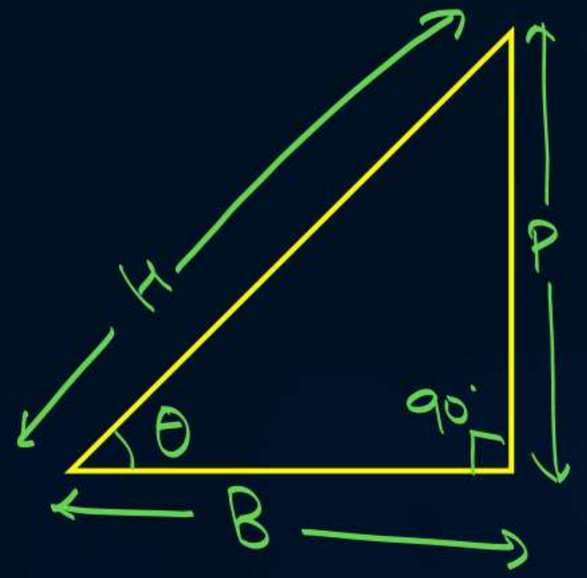
$$H^2 = P^2 + B^2$$

Trigonometric Ratios

1. $\sin\theta = \dfrac{\text{Side opposite to } \angle\theta}{\text{Slant Height}} \Rightarrow \sin\theta = \dfrac{P}{H}$

2. $\cos\theta = \dfrac{\text{Side adjacent to } \angle\theta}{\text{Slant Height}} \Rightarrow \cos\theta = \dfrac{B}{H}$

3. $\tan\theta = \dfrac{P}{B}$



Trigonometry

Fundamental Trigonometric Relations

(1) $\operatorname{Cosec}\theta = \frac{1}{\sin\theta}$ | $\sec\theta = \frac{1}{\cos\theta}$ | $\cot\theta = \frac{1}{\tan\theta}$.

(2) $\tan\theta = \frac{\sin\theta}{\cos\theta}$ | $\cot\theta = \frac{\cos\theta}{\sin\theta}$

(3)
- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \cot^2\theta$
- $1 + \sec^2\theta = \operatorname{Cosec}^2\theta$

Trigonometry

Measurement of an Angle

Degree ($^\circ$)

1° = 60 minutes ($60'$)

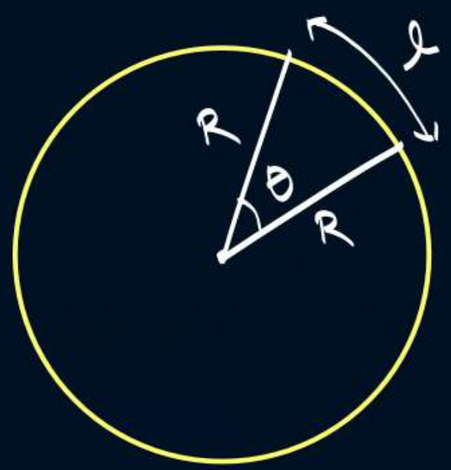
$1'$ = 60 seconds ($60''$)

ex: 45.5°

$= 45^\circ + 0.5^\circ$

$= \boxed{45^\circ 30'}$

Radians



$\Rightarrow \boxed{l = R\theta}$

θ is in Radians.

$\boxed{180^\circ = \pi}$

Degree $\xrightarrow{\times \left(\frac{\pi}{180^\circ}\right)^{\text{radians}}}$ Radians

$\left(\frac{180^\circ}{\pi \text{ radians}}\right) \times$

ex: $45^\circ = (?)$ Radians

$\Rightarrow \cancel{45} \times \frac{\pi}{\cancel{180}\,4} \Rightarrow \frac{\pi}{4}$ Radians.

Circumference:

$2\pi R$

$\theta = 360^\circ$

Trigonometry

$a = 45^\circ$

$\sin(90^\circ) = 1$

$2\sin(45^\circ)\cos(45^\circ)$

$= 2 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = 2 \times \frac{1}{2} = 1$

Double Angle Formulas

$$\sin 2a = 2\sin a \cos a$$

$$\cos 2a = \cos^2 a - \sin^2 a$$
$$= 2\cos^2 a - 1$$
$$= 1 - 2\sin^2 a$$

$$\tan 2a = \frac{2\tan a}{1 - \tan^2 a}$$

Co-function Identities

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$$

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$$

$$\cot\left(\frac{\pi}{2} - \theta\right) = \tan\theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta$$

$$\csc\left(\frac{\pi}{2} - \theta\right) = \sec\theta$$

$$\sec\left(\frac{\pi}{2} - \theta\right) = \csc\theta$$

Even-Odd Identities

$$\sin(-\theta) = -\sin\theta$$

$$\cos(-\theta) = \cos\theta$$

$$\tan(-\theta) = -\tan\theta$$

$$\csc(-\theta) = -\csc\theta$$

$$\sec(-\theta) = \sec\theta$$

$$\cot(-\theta) = -\cot\theta$$

Trigonometry

Half-Angle Formulas

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$$

$$\tan\left(\frac{\theta}{2}\right) = \frac{1-\cos\theta}{\sin\theta}$$

$$= \frac{\sin\theta}{1+\cos\theta}$$

$$= \pm\sqrt{\frac{1-\cos\theta}{2}}$$

Sum-to-Product Formulas

$$\sin a + \sin b = 2\sin\left(\frac{a+b}{2}\right)\cos\left(\frac{a-b}{2}\right)$$

$$\sin a - \sin b = 2\sin\left(\frac{a-b}{2}\right)\cos\left(\frac{a+b}{2}\right)$$

$$\cos a + \cos b = 2\cos\left(\frac{a+b}{2}\right)\cos\left(\frac{a-b}{2}\right)$$

$$\cos a - \cos b = -2\sin\left(\frac{a+b}{2}\right)\cos\left(\frac{a-b}{2}\right)$$

Product-to-Sum Formulas

$$\sin a\sin b = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$$

$$\cos a\cos b = \frac{1}{2}[\cos(a-b) + \cos(a+b)]$$

$$\sin a\cos b = \frac{1}{2}[\sin(a+b) + \sin(a-b)]$$

$$\cos a\sin b = \frac{1}{2}[\sin(a+b) - \sin(a-b)]$$

Trigonometry

Values of Trigonometric Ratios of some Standard Angles

Revision

Ratio \ Angle	0°	30° (π/6)	45° (π/4)	60° (π/3)	90° (π/2)
Sin θ	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
Cos θ	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
Tan θ	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not Defined
Cosec θ	Not Defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
Sec θ	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not Defined
Cot θ	Not Defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

Logarithms

➢ **Definition:**

$\log_b a = x \Rightarrow b^x = a$

$\log_b a = x$ (b: Base, x: exponent)

$\Rightarrow a = b^x$

➢ **Important Properties:**

- $\log(ab) = \log a + \log b$
- $\log\left(\frac{a}{b}\right) = \log a - \log b$
- $\log(a^n) = n\log a$

➢ Base 10 (Common Log) and Base e (Natural Log, ln)

$\log_{10}(24)$

$\log_e(24) = \ln(24)$

Logarithms

Basic Identities

1.Log of a product:

$$\log_b(xy) = \log_b(x) + \log_b(y)$$

2.Log of a quotient:

$$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$$

3.Log of a power:

$$\log_b(x^n) = n \cdot \log_b(x)$$

$\log_{10}(100) = 2$
$\log_{10}(1000) = 3$
} only for Base = 10 (Common Log.)

4.Log of 1:

$$\log_b(1) = 0 \qquad \text{(Because } b^0 = 1\text{)}$$

5.Log of the base itself: ex: $\log_{10^7}(10^7) = 1$

$$\log_b(b) = 1 \qquad \text{(Because } b^1 = b\text{)}$$

6.Log of base raised to log:

$$b^{\log_b(x)} = x$$

ex: $(243)^{\log_{243}(2)} = 2$

Logarithms

Natural Logarithm (base e)

- $\ln(e) = 1$ $\quad [\log_e(e) = 1]$
- $\ln(1) = 0$
- $\ln(xy) = \ln(x) + \ln(y)$
- $\ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$
- $\ln(x^n) = n \cdot \ln(x)$

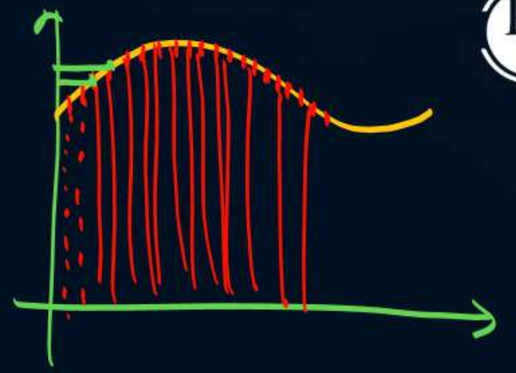
$\log_{10}(2) = ?$
$\log_{10}(3) = ?$
$\log_{10}(5) = ?$
→ learn.

Que → Simplify :- [H.W.]

$$\log_2\left(10^{\log_{10}(1024)}\right)$$

A) 1
B) 0
C) 10
D) 100

Differentiation (Derivatives)



Slope = $\frac{\Delta y}{\Delta x}$

Differentiation = Finding the rate of change

- If $y = f(x)$, then $\frac{dy}{dx}$ tells how y changes as x changes.
- **In Physics:**

QUANTITY	DERIVATIVE
Velocity	$v = \frac{ds}{dt}$
Acceleration	$a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$
Force (from energy)	$F = -\frac{dU}{dx}$

Double derivative.

Differentiation (Derivatives)

$y = f(x)$

$y' = \frac{dy}{dx}$ | $y'' = \frac{d^2y}{dx^2}$

Derivative | Double Derivative.

- **Basic Rules:**
- If $f(x) = x^n$, then: $\frac{d}{dx}(x^n) = nx^{n-1}$

$y = f(x)$

$y = x^n$

$\frac{dy}{dx} = nx^{n-1}$

Derivative of a constant term is always zero.

Function	Derivative
x^n	nx^{n-1}
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
★★ e^x	e^x
$\ln x$	$\frac{1}{x}$

Valid for both, natural and common log.

Differentiation (Derivatives)

Product Rule

$$y(x) = u(x) \cdot v(x)$$

$$\Rightarrow \boxed{y'(x) = u'(x) \cdot v(x) + u(x) \cdot v'(x)}$$

ex: $y = \sin x \cdot \cos x$

By Product Rule :—

$$\frac{dy}{dx} = \left[\frac{d}{dx}(\sin x)\right]\cos x + \sin x\left[\frac{d}{dx}(\cos x)\right]$$

$$\Rightarrow y' = \cos x \cdot \cos x + \sin x \cdot (-\sin x) \Rightarrow \boxed{\frac{dy}{dx} = \cos^2 x - \sin^2 x}$$

ex: $y = \left(\frac{3}{2}x^3 + 4x\right)\cdot\left(5x^2 + 6\right)$

$$\frac{dy}{dx} = \left(3 \times \frac{3}{2}x^2 + 4\right)\left(5x^2 + 6\right) + \left(\frac{3}{2}x^3 + 4x\right)\left[2 \times 5x + 0\right]$$

$$\Rightarrow y' = \left(\frac{9}{2}x^2 + 4\right)\left(5x^2 + 6\right) + \left(\frac{3}{2}x^2 + 4x\right)10x$$

H.W.

Que: $y = \log(x) \cdot e^x$

Find — (i) $\frac{dy}{dx}$

(ii) $\frac{d^2y}{dx^2}$

Differentiation (Derivatives)

Quotient Rule

$$y(x) = \frac{u(x)}{v(x)}$$

$$\Rightarrow \boxed{y'(x) = \frac{u'(x)\cdot v(x) - u(x)\cdot v'(x)}{[v(x)]^2}}$$

ex: $y = \frac{\sin x}{\cos x}$ } $\tan x$

$$y' = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{(\cos x)^2}$$

$$y' = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$\boxed{y' = \sec^2 x}$$

Derivative of $\tan x$.

ex: $y = \dfrac{(\log x)}{e^x}$ | Find — (i) y' (ii) y''

Sol^n (i) $y' = \dfrac{\left(\frac{1}{x}\right)e^x - (\log x)e^x}{(e^x)^2}$

$\Rightarrow y' = \dfrac{e^x\left[\frac{1}{x} - \log x\right]}{e^{2x}\; e^x}$

$\Rightarrow \boxed{y' = \dfrac{\left(\frac{1}{x} - \log x\right)}{(e^x)}}$

$y' = \dfrac{(x^{-1} - \log x)}{e^x}$

(ii) $y'' = \dfrac{(-x^{-2} - x^{-1})\cdot e^x - (x^{-1} - \log x)e^x}{e^{2x}}$

$\Rightarrow y'' = \dfrac{\left[\left(\frac{-1}{x^2} - \frac{1}{x}\right) - \frac{1}{x} + \log x\right]e^x}{e^{2x}\; e^x}$

$\Rightarrow \boxed{y'' = \dfrac{\log x - \frac{2}{x} - \frac{1}{x^2}}{e^x}}$

Differentiation (Derivatives)

Chain Rule → बाहर से अंदर तक जाना है।

$y = \log(x^2 + \sin^3 x)$ | Find $\frac{dy}{dx}$ for $x = \pi$.

$$\Rightarrow y' = \left(\frac{1}{x^2 + \sin^3 x}\right)\left(2x + (3\cdot\sin^2 x)\cdot\cos x\right)$$

$$\Rightarrow y' = \frac{2x + 3\sin^2 x\cdot\cos x}{x^2 + \sin^3 x}$$

$$\Rightarrow y' = \frac{2(\pi) + 3\sin^2(\pi)\cdot\cos(\pi)}{(\pi)^2 + \sin^3(\pi)} \Rightarrow y' = \frac{2\pi}{\pi^2} \Rightarrow \boxed{y' = \frac{2}{\pi}}$$

Que: $\log\left(\sin\left(e^{3x^2+4}\right) + \cos\left(e^{x}\right)\right)$

Find $\frac{dy}{dx} = ?$.

Hint:

$e^x \xrightarrow{\text{derivative}} e^x$

$e^{(\cdots)} \xrightarrow{\quad=\quad} e^{(\cdots)}$

Integration (Anti-Derivatives)

$v = \frac{ds}{dt}$



- **Integration = Adding up small quantities to get the total**

- **In Physics:**

Quantity	Integration
Displacement	$s = \int v dt$
Velocity	$v = \int a dt$
Work Done	$W = \int F dx$

Integration (Anti-Derivatives)

$y = 3x^2 + 2x + 5$

$y' = 6x + 2$

$\int 6x\,dx = \frac{6x^2}{2} + 2x + C$

$= 3x^2 + 2x + C$

- Integration = Adding up small quantities to get the total
- Note : C is the constant of integration
- Basic Formulas:

$x^{-1} = \frac{1}{x}$

Function	Integral
$\int x^n dx$	$\frac{x^{n+1}}{n+1} + C, n \neq -1$
$\int \sin x dx$	$-\cos x + C$
$\int \cos x dx$	$\sin x + C$
$\int e^x dx$	$e^x + C$
$\int \frac{1}{x} dx$	$\ln x + C$

ex: $y = \int (x^3 + 5x^2 + 6x + 7)\, dx$

$= \frac{x^4}{4} + 5\frac{x^3}{3} + \overset{3}{\cancel{6}}\frac{x^2}{\cancel{2}} + 7x + C$

ex: $y = \int e^{(3x+4)} \cdot dx$

$= \frac{e^{3x+4}}{3} + c$

H.W.

(i) $y = \int \log(\sin x^2) \cdot dx$

(ii) $y = \int \sin(\cos x)\, dx$

Integration (Anti-Derivatives)

Indefinite → variable value

Definite Integral

↳ We have defined limits.

$$y = \int_{x=x_1}^{x=x_2} x^n \, dx = \left[\frac{x^{n+1}}{n+1} \right]_{x=x_1}^{x=x_2} = \left(\frac{x_2^{n+1}}{n+1} - \frac{x_1^{n+1}}{n+1} \right) + c$$

$$\Rightarrow y = \frac{x_2^{n+1} - x_1^{n+1}}{n+1} + c$$

Ex: $y = \int_{x=1}^{x=3} (3x^2 + 4)\,dx$

$\Rightarrow y = \left[\frac{3x^3}{3} + 4x\right]_{x=1}^{x=3}$

$\Rightarrow y = (27 + 12) - (1 + 4)$

$\Rightarrow \boxed{y = 34 + C}$

H.W.

Ques: $y = \int_{x=0}^{x=2} (5x^3 + 6x^2 + 2x + 7)\,dx$

If the value of y is 27, then find the value of integration constant C.

Some Important Curves

Straight Line

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{y_1 - y_2}{x_1 - x_2}$$

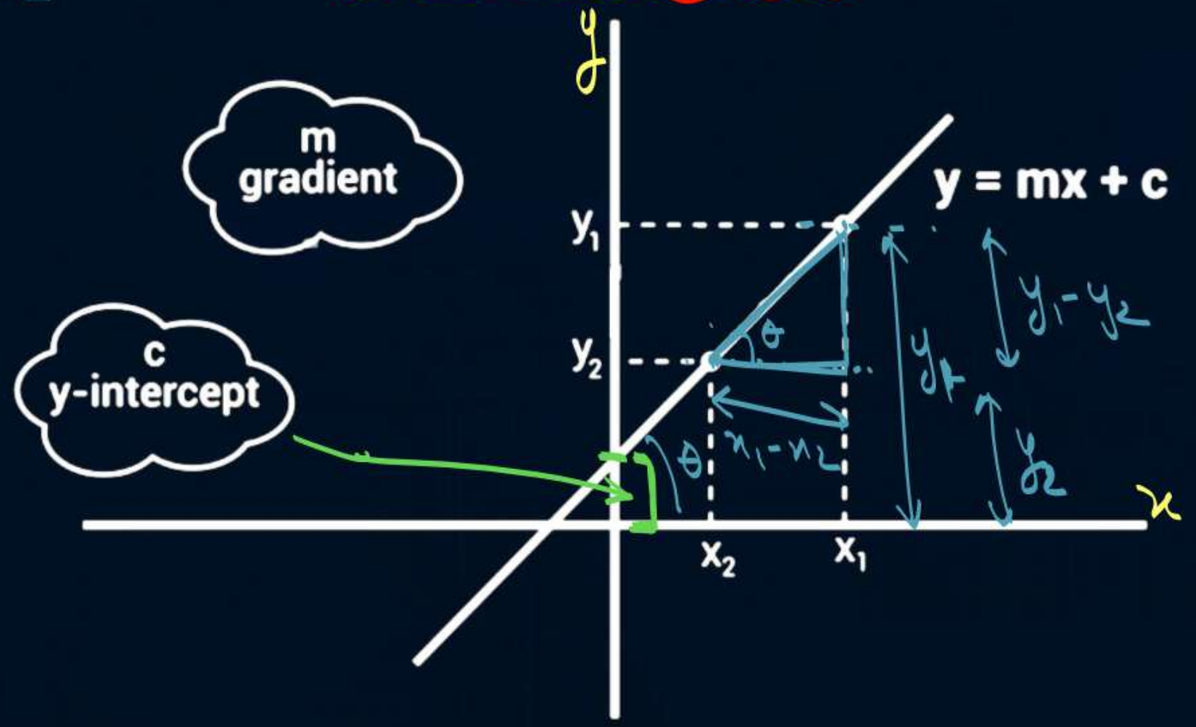
$$\tan\theta = \frac{P}{B} = \frac{y_1 - y_2}{x_1 - x_2}$$

$$m = \frac{\Delta y}{\Delta x}$$

$$m = \tan\theta$$

Slope — y-intercept

$$y = mx + c$$

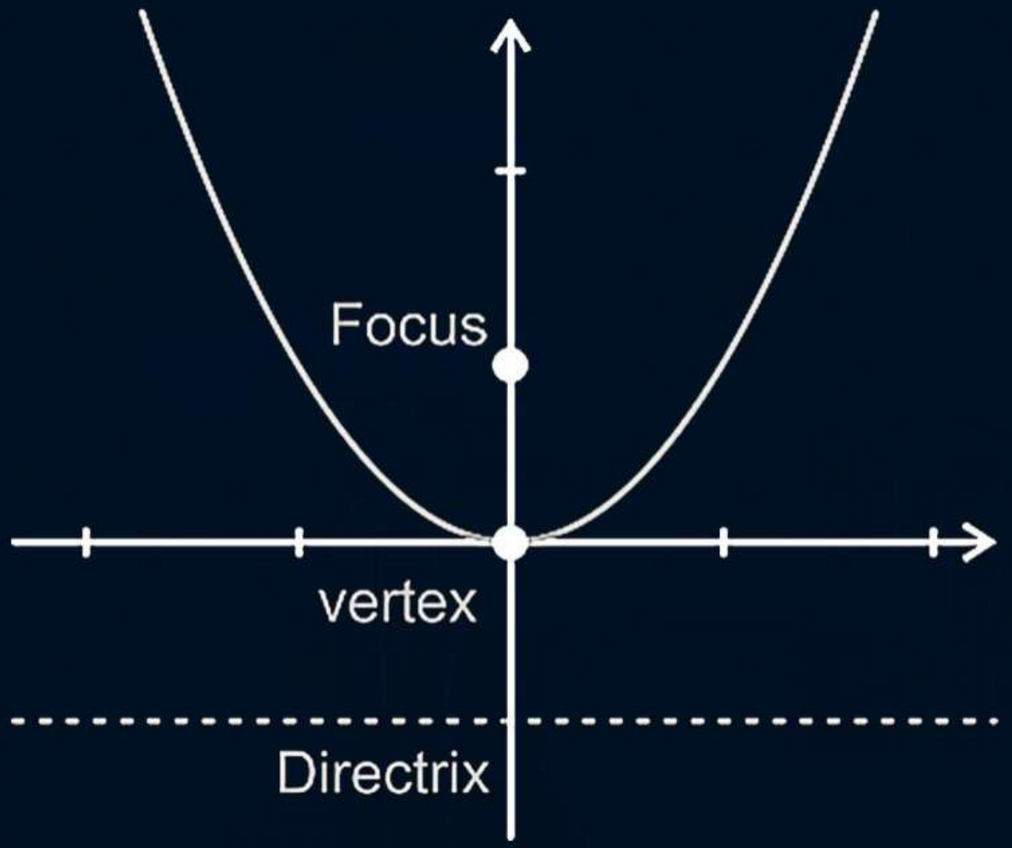


Area under the curve

Some Important Curves

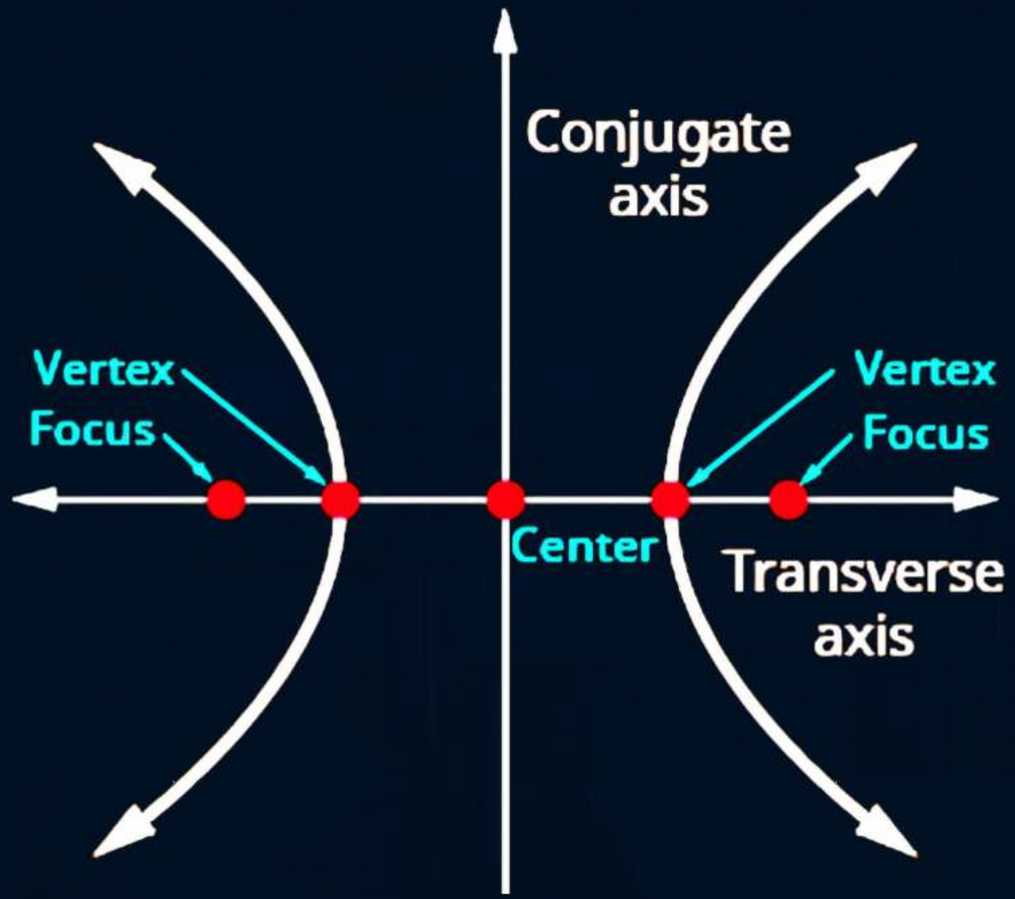
Parabola

ex: Projectile.



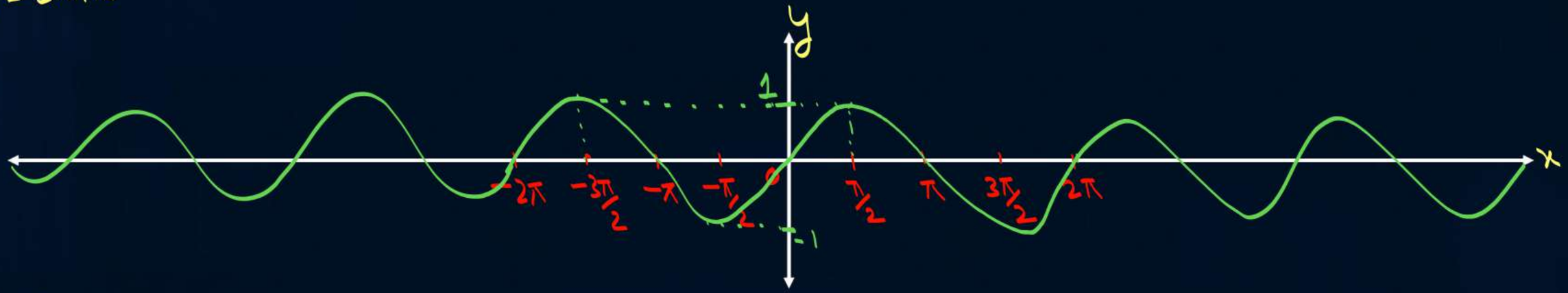
Some Important Curves

Hyperbola

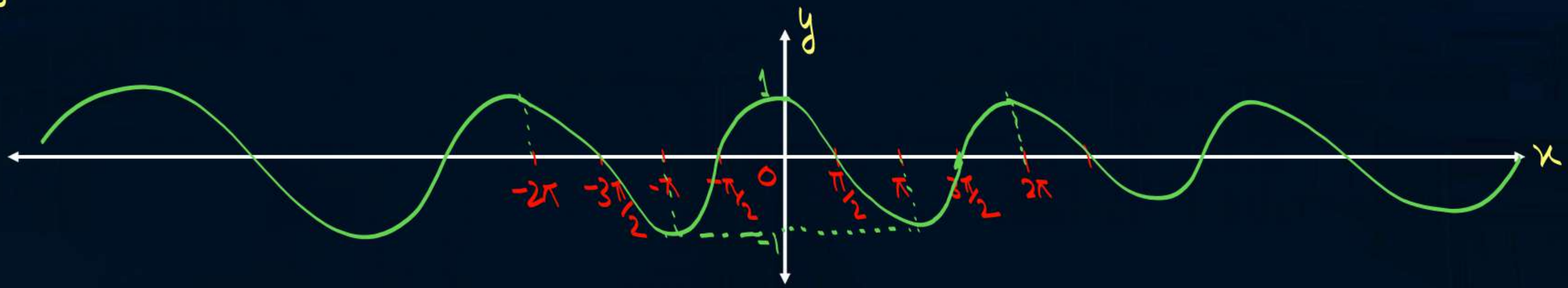


Some Important Curves

$y = \sin x$

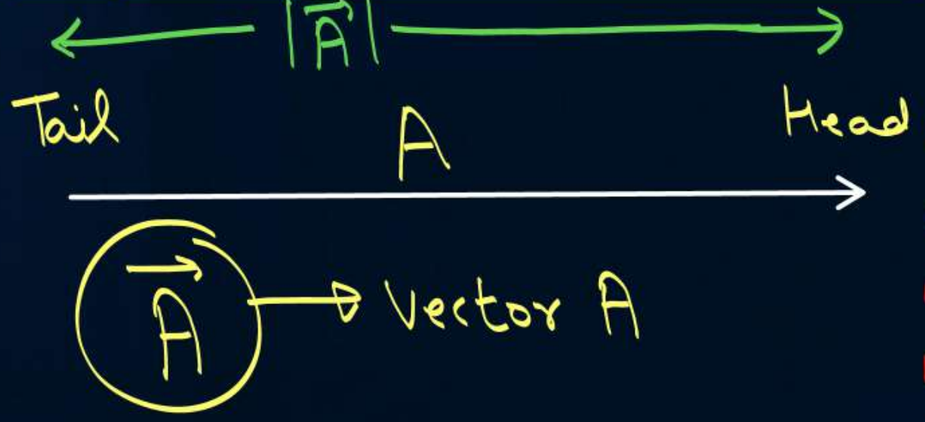


$y = \cos x$



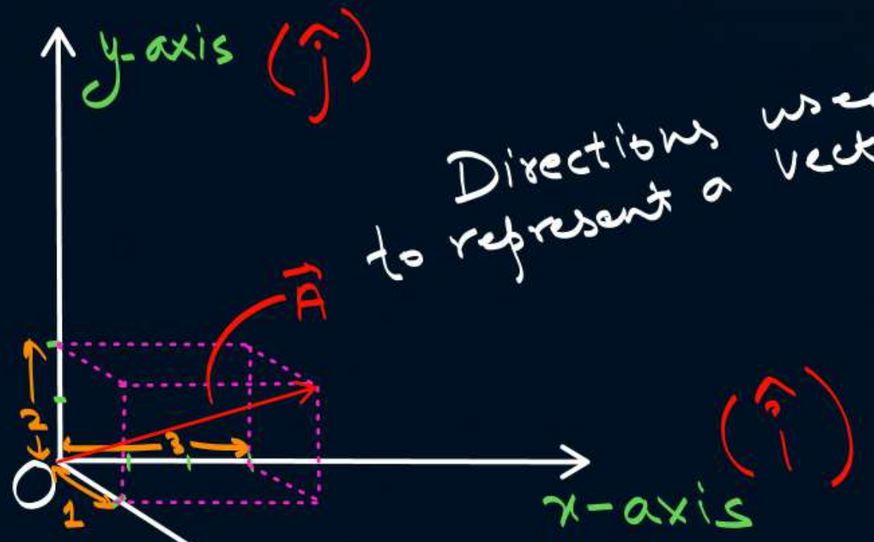
Vectors

Representation of a vector



$\vec{A}$ → Vector A

$|\vec{A}|$ → Magnitude of $\vec{A}$ (value)



ex: $\vec{A} = 3\hat{i} + 2\hat{j} + \hat{k}$

- $3\hat{i}$: x-component of $\vec{A}$
- $2\hat{j}$: y-component of $\vec{A}$
- $\hat{k}$: z-component of $\vec{A}$

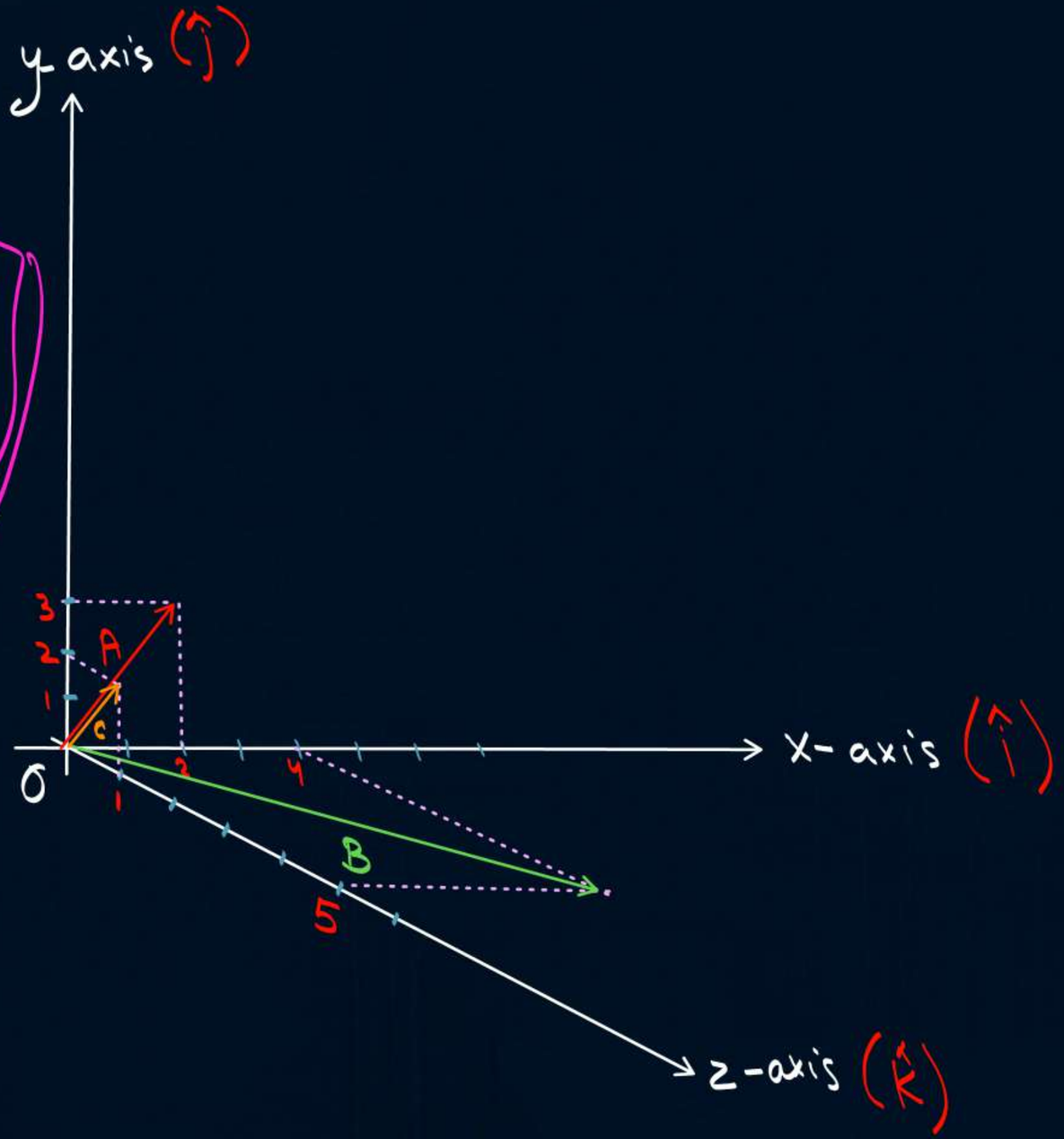
3D vector

$\vec{A} = 2\hat{i} + 3\hat{j}$ (x-y plane)

$\vec{B} = 4\hat{i} + 5\hat{k}$ (x-z plane)

$\vec{C} = 2\hat{j} + \hat{k}$ (y-z plane)

2D Vectors



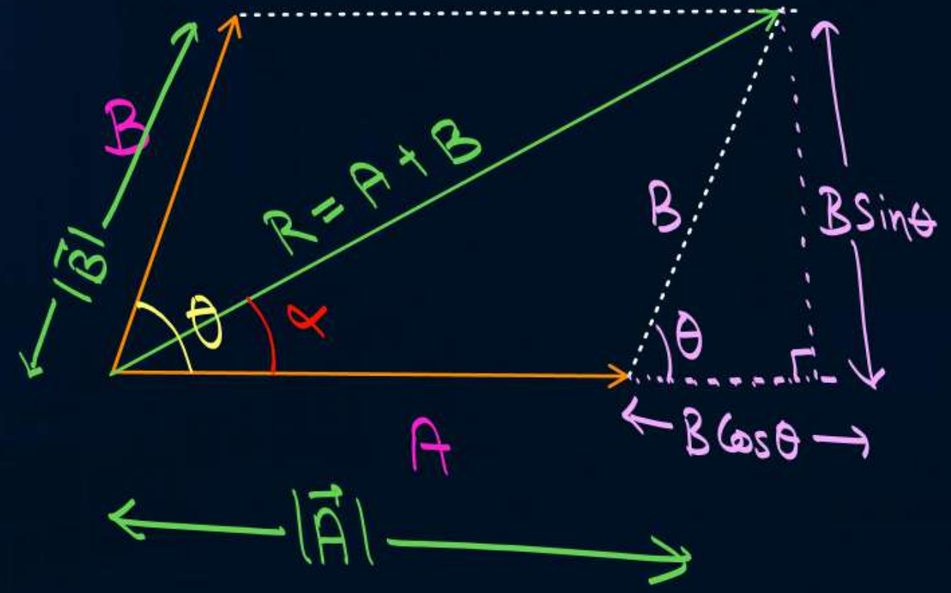
Vectors

Resultant of two vectors

$$|\vec{R}| = \sqrt{|\vec{A}|^2 + |\vec{B}|^2 + 2|\vec{A}||\vec{B}|\cos\theta}$$

↑ Magnitude of the Resultant vector of $\vec{A}$ and $\vec{B}$.

$$\vec{R} = \vec{A} + \vec{B}$$



Direction of Resultant vector :-

$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$

$$\Rightarrow \alpha = \tan^{-1}\left(\frac{B\sin\theta}{A + B\cos\theta}\right)$$

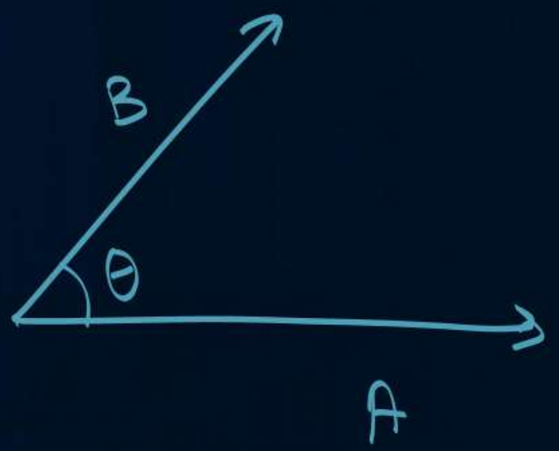
Vectors

Dot Product or Scalar Product

Dot Product of $\vec{A}$ and $\vec{B}$ $\Rightarrow$

$$\vec{A}\cdot\vec{B} = |\vec{A}||\vec{B}|\cos\theta$$

Scalar Quantity.



ex: $\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k}$

$\vec{B} = 5\hat{i} + 2\hat{j} + 7\hat{k}$

$$\Rightarrow \cos\theta = \frac{\vec{A}\cdot\vec{B}}{|\vec{A}||\vec{B}|} = \frac{10+6+28}{\sqrt{29}\sqrt{78}}$$

$$|\vec{A}| = \sqrt{2^2+3^2+4^2} = \sqrt{29}$$

$$|\vec{B}| = \sqrt{5^2+2^2+7^2} = \sqrt{78}$$

$$\Rightarrow \cos\theta = \frac{44}{\sqrt{29\times 78}}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{44}{\sqrt{2262}}\right)$$

Note

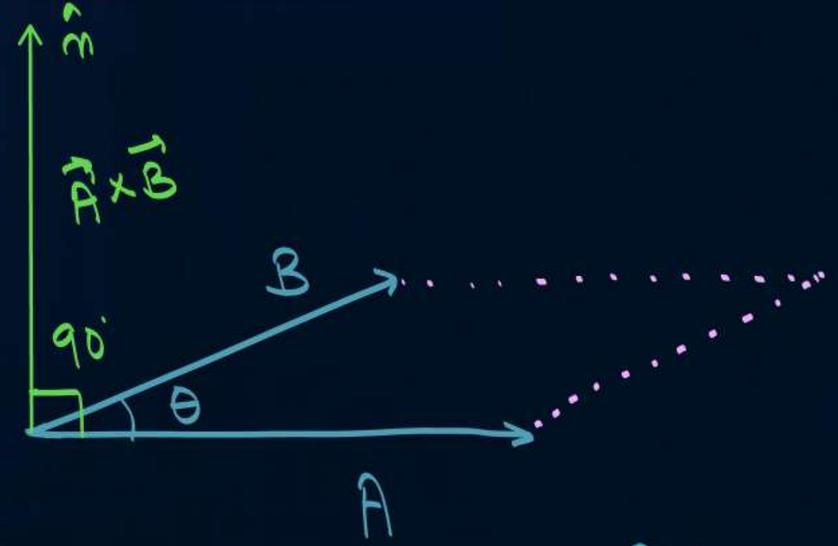
जब भी 2 vectors दिये हो और उनके बीच का angle find करना हो, तब Dot Product use करो।

Vectors

Cross Product or Vector Product

$$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}| \sin\theta \, \hat{n}$$

$\hat{n}$ → unit vector perpendicular to the plane in which both the vectors $\vec{A}$ and $\vec{B}$ are lying.



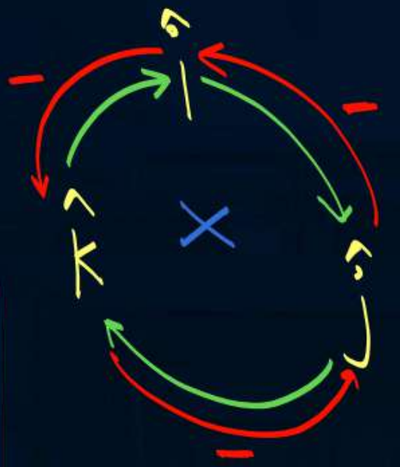
ex: $\vec{A} = 2\hat{i} + \hat{j} + 2\hat{k}$

$\vec{B} = -\hat{i} + 3\hat{j}$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 2 \\ -1 & 3 & 0 \end{vmatrix}$$

$$= \hat{i}(1\times 0 - 3\times 2) - \hat{j}(2\times 0 - (-1)\times 2) + \hat{k}(2\times 3 - (-1)\times 1)$$

$$\Rightarrow \vec{A} \times \vec{B} = -6\hat{i} + 2\hat{j} + 5\hat{k}$$



Basic Math & Calculus

THANK YOU

